

09

GRAPH TRAVERSAL

9.1 Introduction

9.1.1 Breadth First Search

- Uses a queue data structure.
- To find whether a graph is connected or not.
- To find number of connected components in a given graph (Breadth First traversal is used)
- To detect cycle in a graph.
- To find whether a given graph is bipartite or not.
- T.C. = $O(V + E)$ using Adjacency List
= $O(V^2)$ using Adjacency matrix.
- To find shortest path in undirected graph without weights.

9.1.2 Depth First Search

- Uses stack.
- To detect a cycle in a graph.
- To find whether a graph is connected or not.
- To find whether given graph is bipartite or not.
- To find number of connected components in the graph.
- To find bridges in the graph.
- To find articulation points in the graph.
- To find topological sort of a graph.
- To find biconnected components.
- To find strongly connected components.

